PREDICTING HOSPTITAL STAYS

#bY @TOBIAS @MAXIMILLIAN @RAGHVENDRA @SOURABH

Content

Overview

INTRODUCTION

PROBLEM STATEMENTS

DATA CLEANING AND INTERPRETATION

VISUALS AND STATS

ANALYSIS AND TESTING

SUMMARY AND CONCLUSION

DATA SET

case_id	Hospital_code	Hospital_type_code	City_Code_Hospital	Hospital_region_code	Available Extra Rooms in Hospital	Department	Ward_Type	Ward_Facility_Code
1 to 318438	1 to 32	a, b, c, d, e, f, g	1 to 13	X, Y, Z	0 to 24	TB & Chest disease, anesthesia, gynecology, radiotherapy, Surgery	P, Q, R, S, T, U	A, B, C, D, E, F

Bed Grade	patientid	City_Code_Patient	Type of Admission	Severity of Illness	Visitors with Patient	Age	Admission_Deposit	Stay
1 to 4	1 to 92017	1 to 37	Emergency, Urgent, Trauma	Extreme, Moderate, Minor	0 to 32	'0-10' to '91-100'	1800 to 11008	'0-10' to '91-100' and 'More than 100'

DATA DICTIONARY

case_id	Hospital_code	Hospital_type_code	City_Code_Hospital	Hospital_region_code	Available Extra Rooms in Hospital	Department	Ward_Type	Ward_Facility_Code
Case_ID registered in Hospital	Unique code for the Hospital	Unique code for the type of Hospital	City Code of the Hospital	Region Code of the Hospital	Number of Extra rooms available in the Hospital	Department overlooking the case	Code for the Ward type	Code for the Ward

Bed Grade	patientid	City_Code_Patient	Type of Admission	Severity of Illness	Visitors with Patient	Age	Admission_Deposit	Stay
Condition of Bed in Ward	Unique Patient Id	City Code for the patient	Admission Type registered by the Hospital	Severity of the illness recorded at the time of admission	Number of Visitors with the patient	Age of the patient	Deposit at the Admission Time	Stay Days by the patient

FIRST IMPRESSIONS

THE GOOD

- easy data cleaning
- many interesting variables
- plentiful data

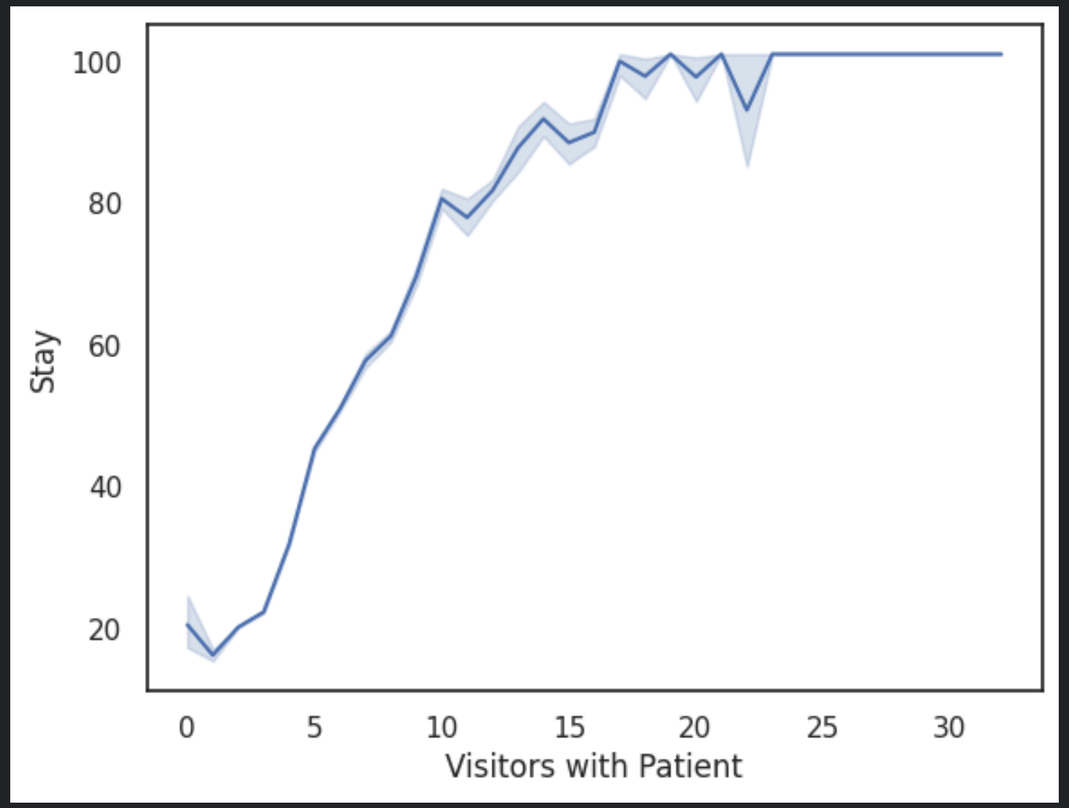
THE BAD

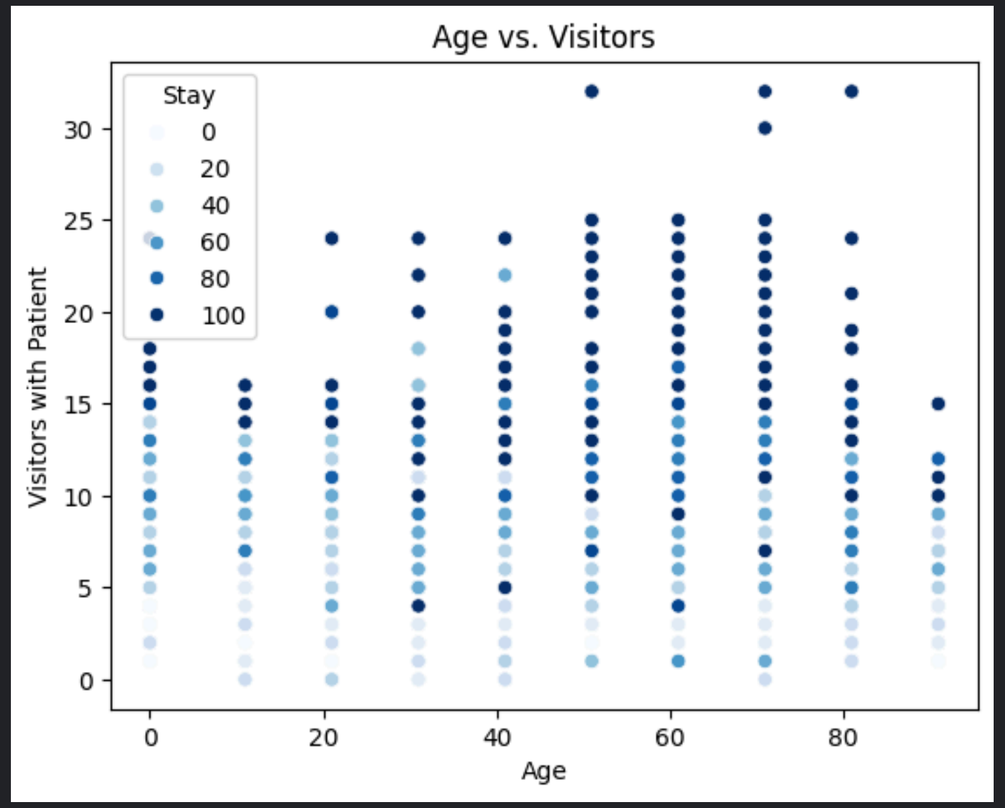
- useless data dictionary resulting in
- some useless variables
- obscured 'Age' and 'Stay' variables through categorical '0-10' encoding

Problems

Problems

Problems





INTERESTING OBSERVATION📉

DATA CLEANING AND INTERPRETATION

Why Data Cleaning Matters.....

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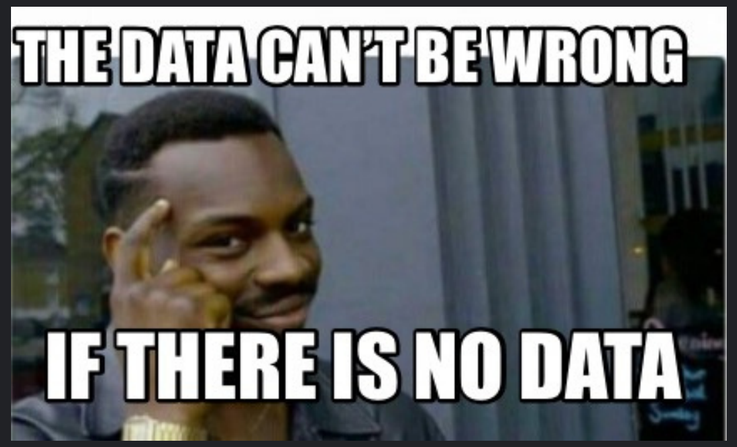
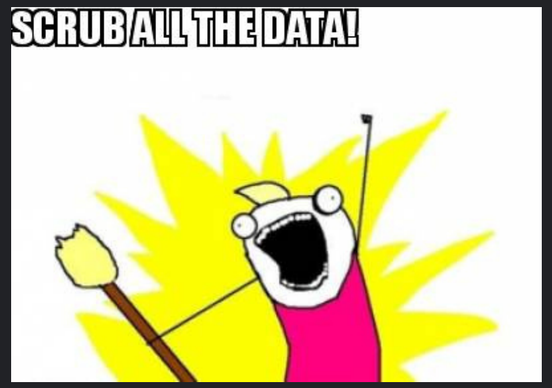


Initial Data Quality Issues....

Initial Data Quality Issues



Data Cleaning Steps

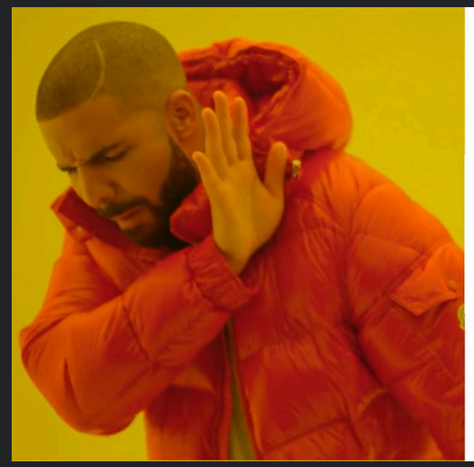
```python
# Impute City_Code_Patient
df_clean['City_Code_Patient'] = df_clean['City_Code_Patient'].fillna(-1)

# Drop rows with missing Bed Grade and Stay
df_clean.dropna(subset=['Bed Grade', 'Stay'], inplace=True)
```

```python
    value = str(value).strip()
    if "More than" in value:
        return 110
    if '-' in value:
        try:
            start, end = map(int, value.split(
            return (start + end) / 2
        except:
            return None
```

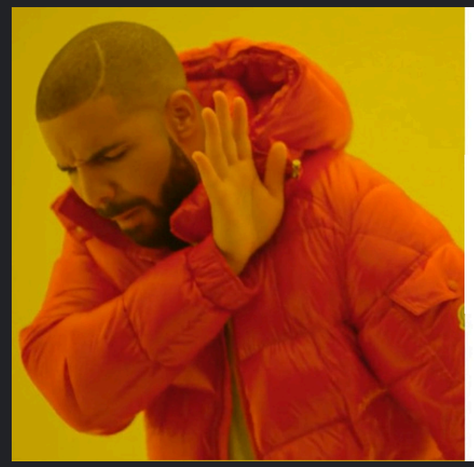
Cleaning Outcomes

Cleaning Outcomes



```
🌸 Missing Values Before Cleaning:
Bed Grade                148
City_Code_Patient       6689
Stay                  137057
dtype: int64
```

Cleaning Outcomes



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🌸 Missing Values Before Cleaning:
Bed Grade                148
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Stay                  137057
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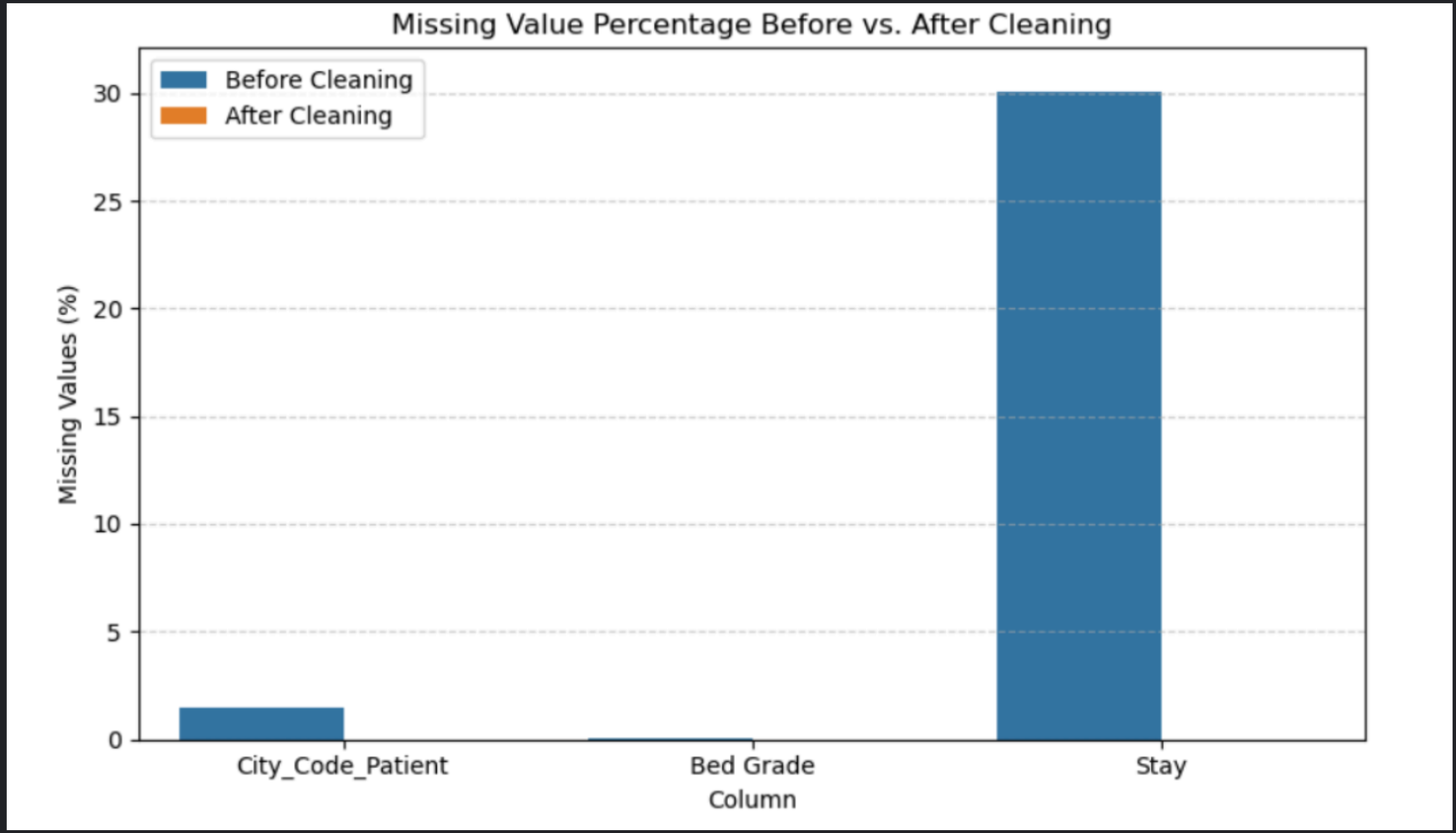
```

```
✅ Missing Values After Cleaning:
Series([], dtype: int64)
```

Visualizing Impact of Cleaning

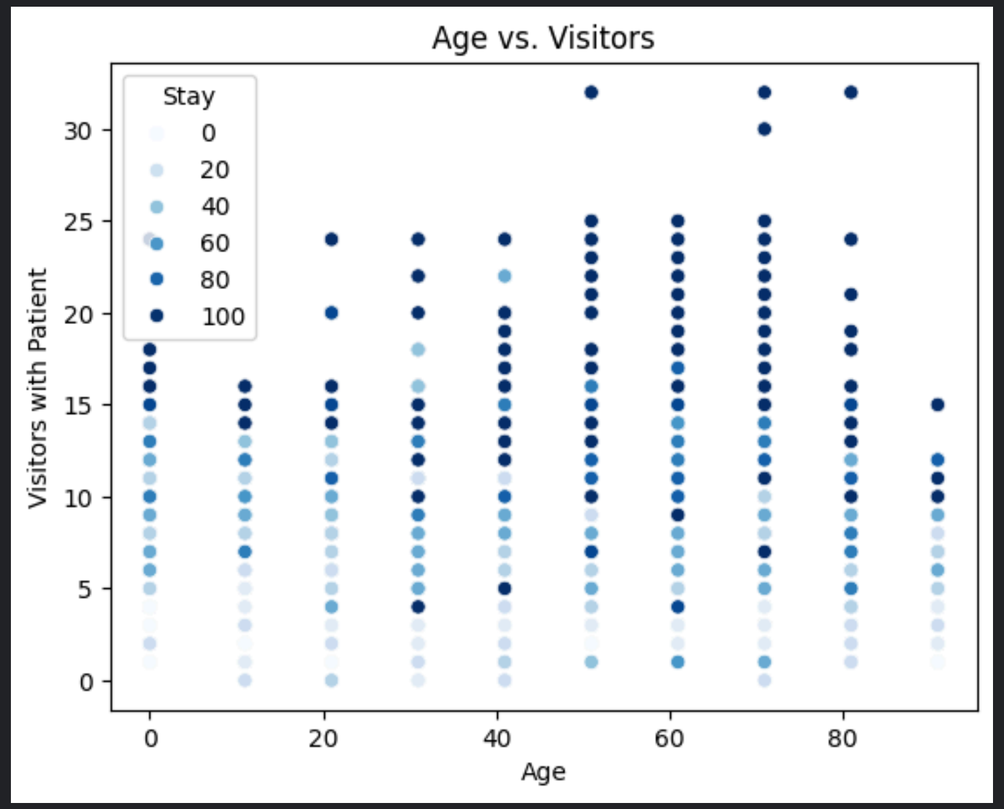


VISUALS, GRAPHS AND STATS

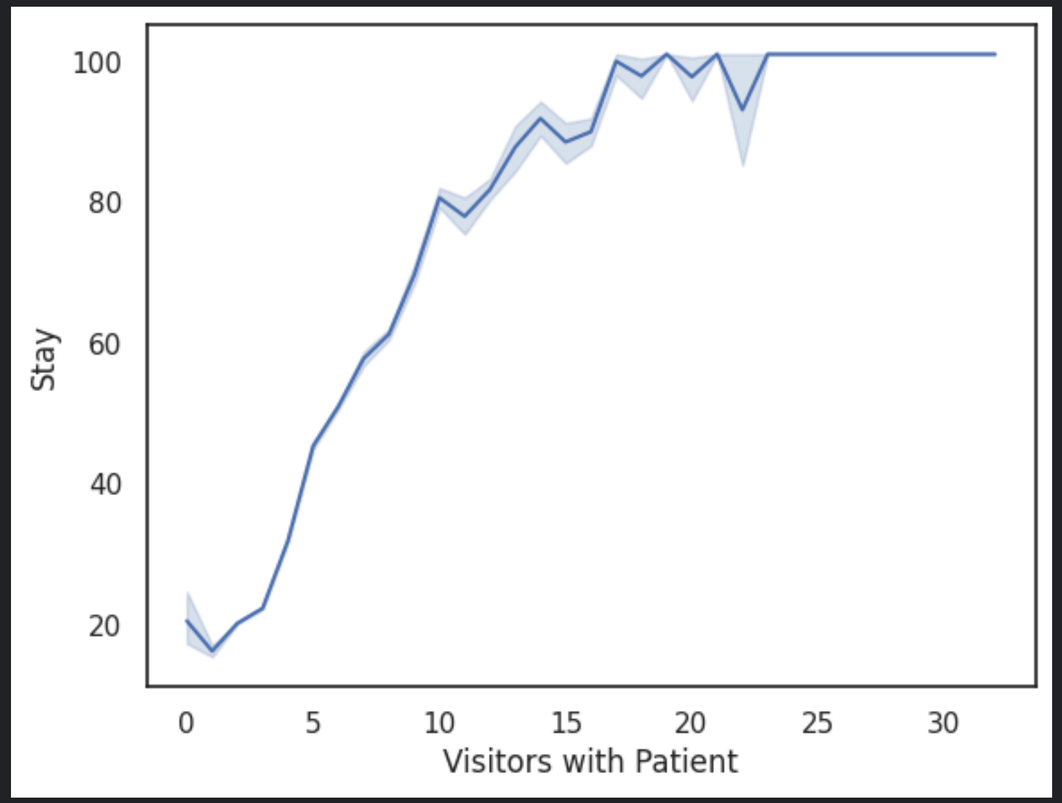
Visualisation of relevant data relationships

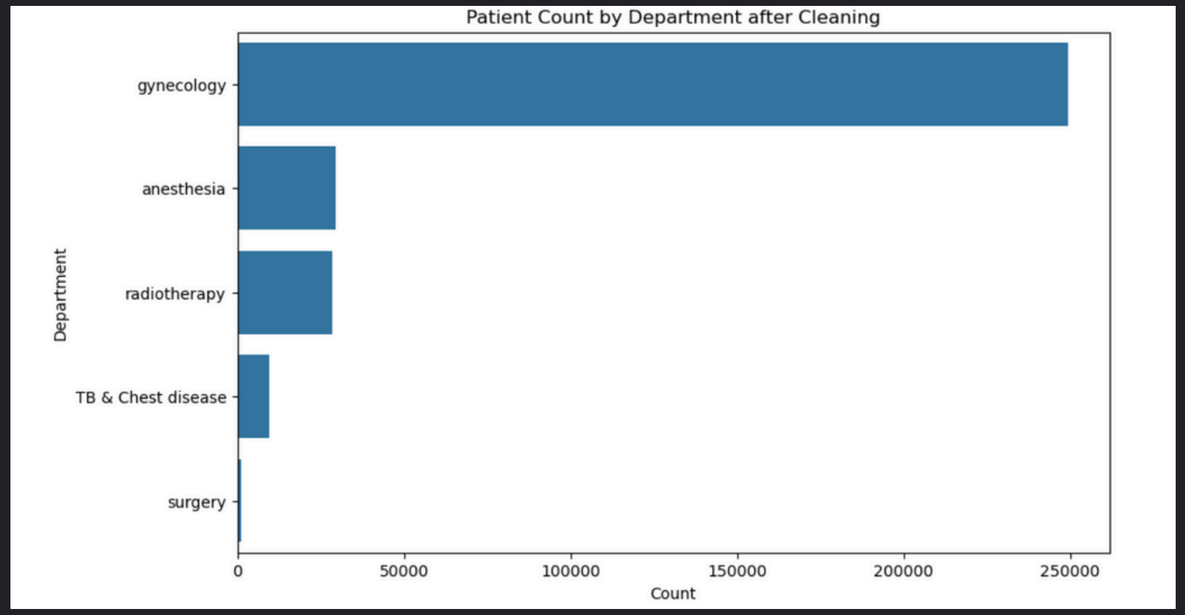
-It was important to include graphs that were easily recognizable
-The code itself generates automatically based on the data given
-Relevant relationships: Age, number of visitors, length of stay
-Type of graphs used: scatter plot and line graph

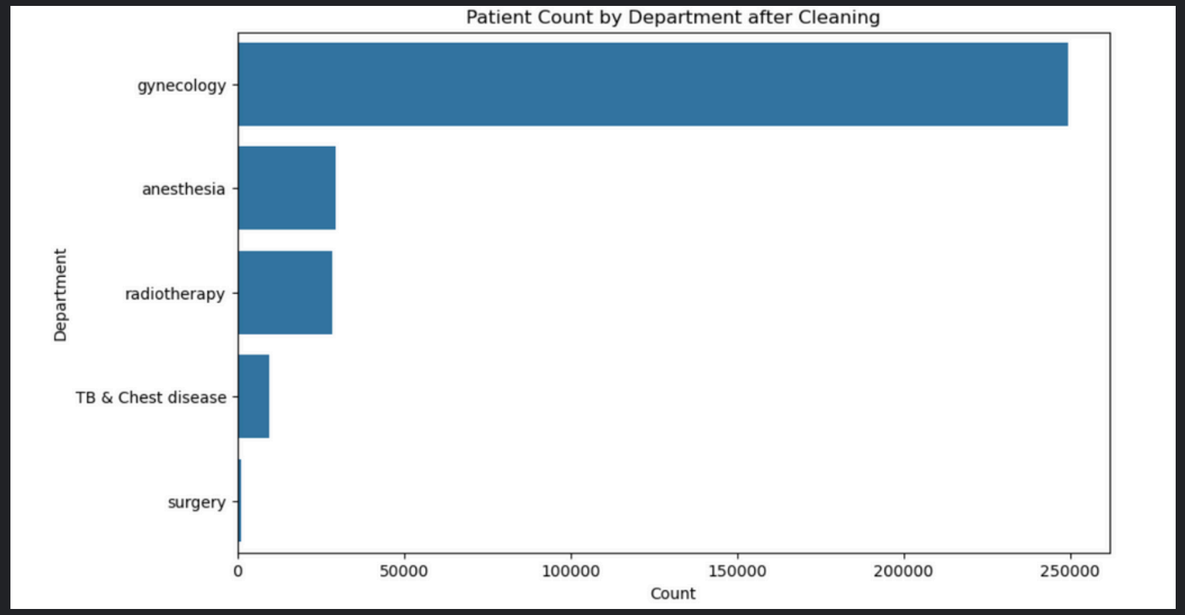
Scatter plot



Line graph




Patient Count by Department after Cleaning
gynecology
anesthesia
radiotherapy
TB & Chest disease
surgery
Department
0
50000
100000
150000
200000
250000
Count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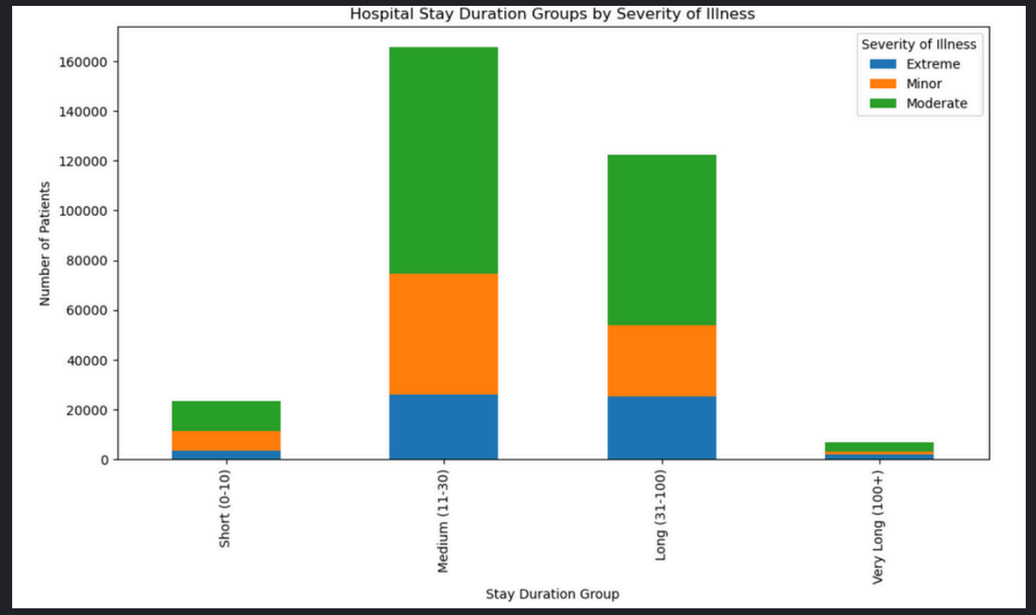
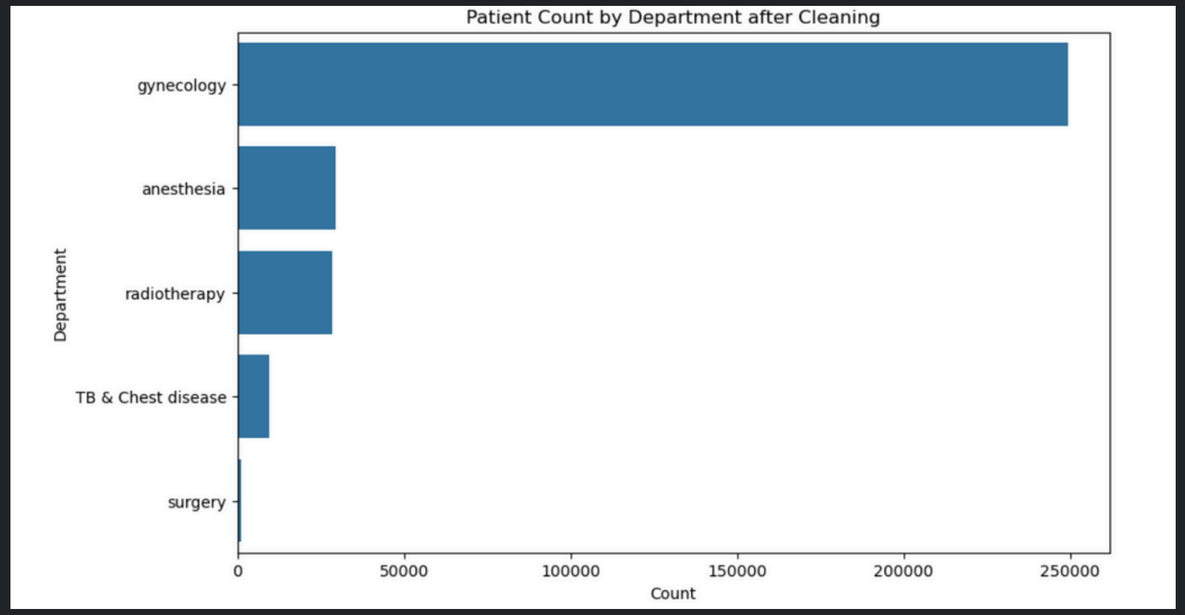

Hospital Stay Duration Groups by Severity of Illness
Severity of Illness
Extreme
Minor
Moderate
Number of Patients
0
20000
40000
60000
80000
100000
120000
140000
160000
Short (0-10)
Medium (11-30)
Long (31-100)
Very Long (100+)
Stay Duration Group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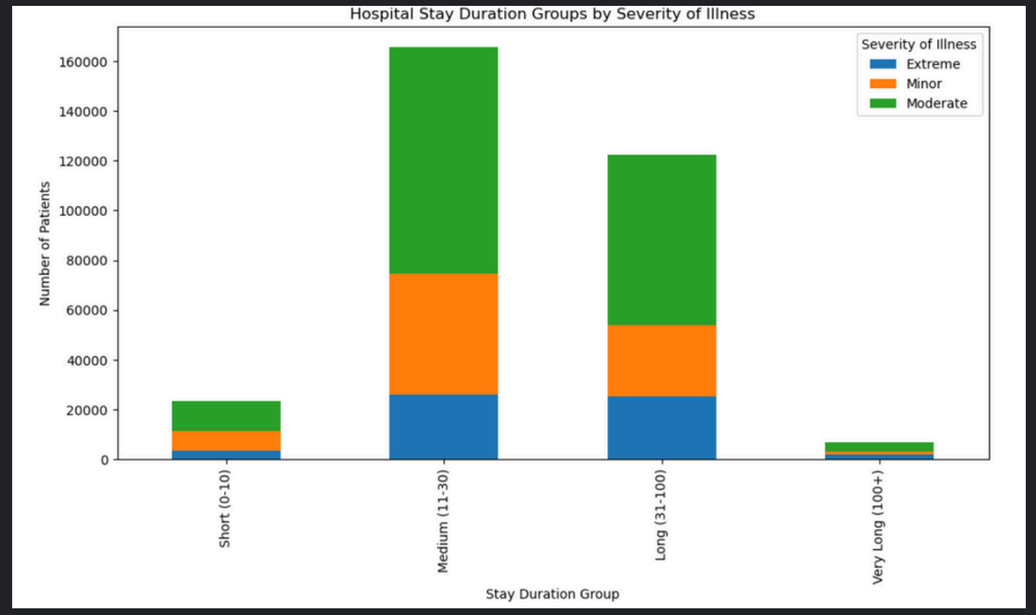
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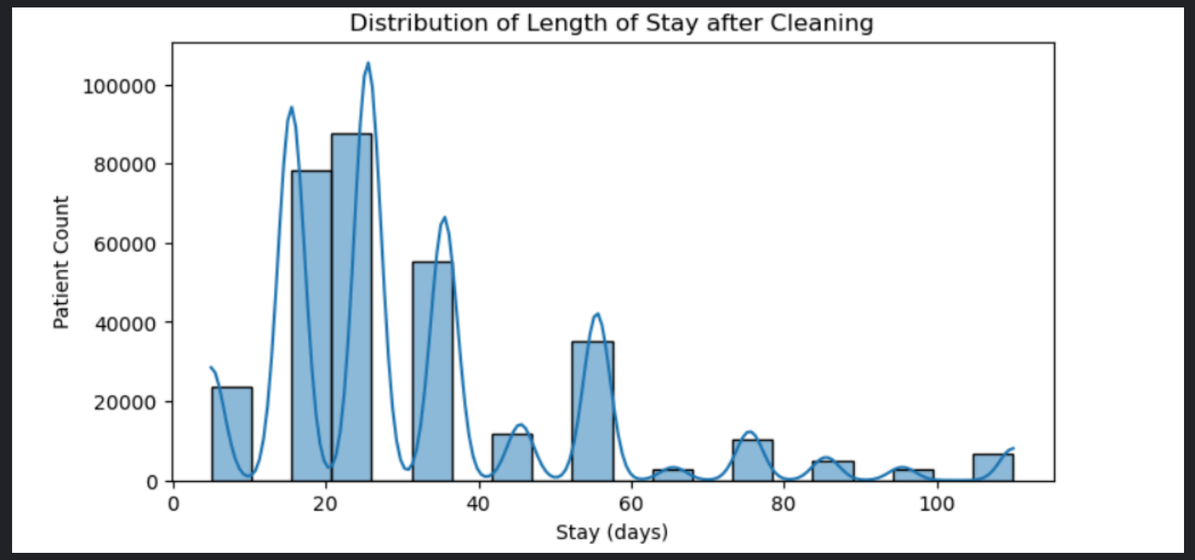
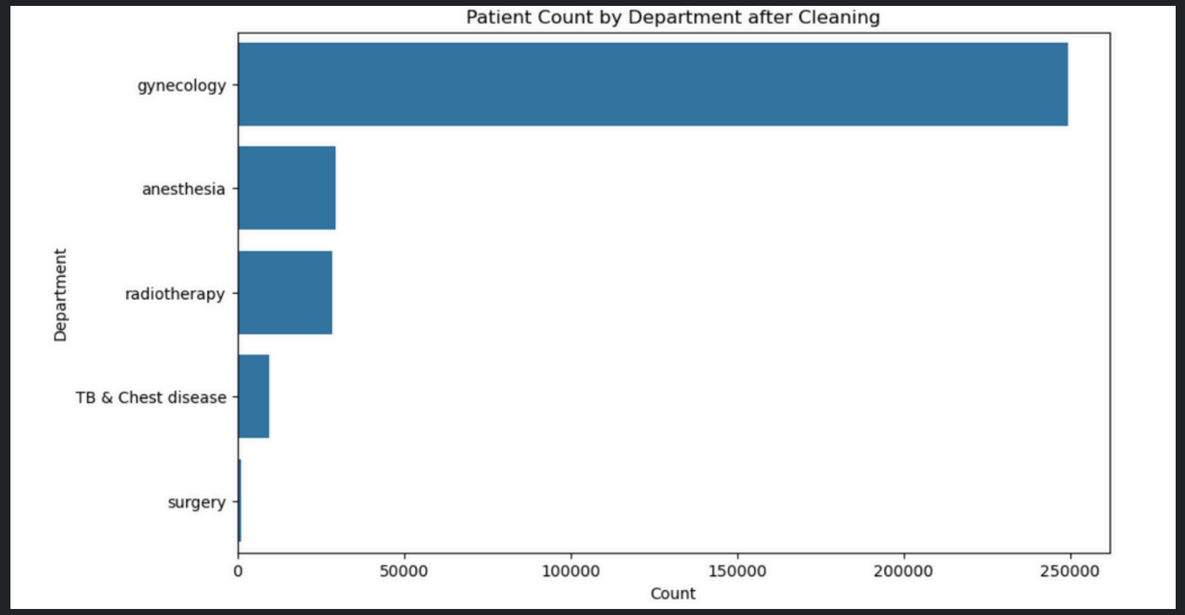

Distribution of Length of Stay after Cleaning
Patient Count
0
20000
40000
60000
80000
100000
0
20
40
60
80
100
Stay (days)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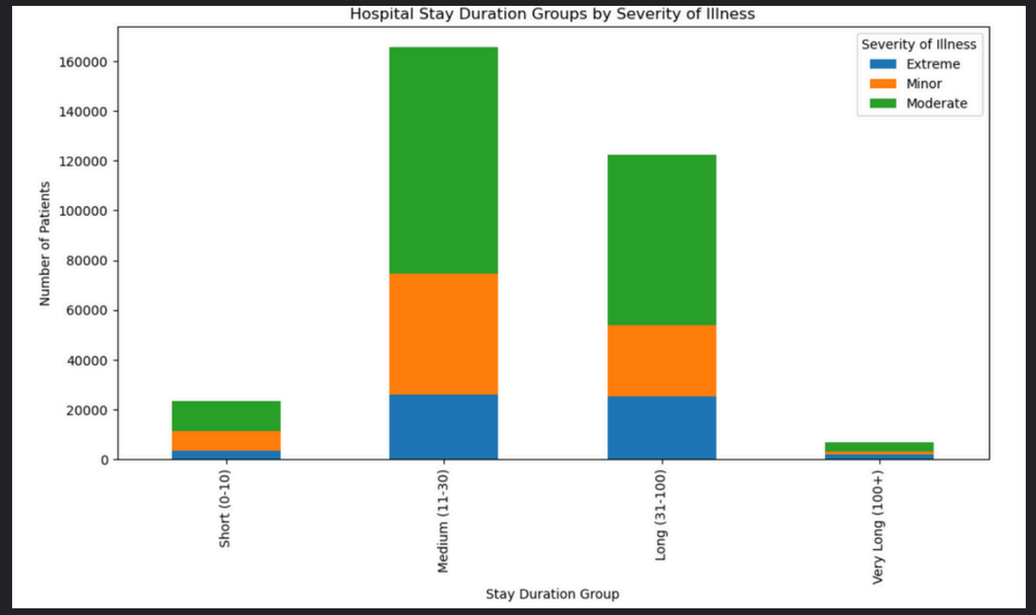
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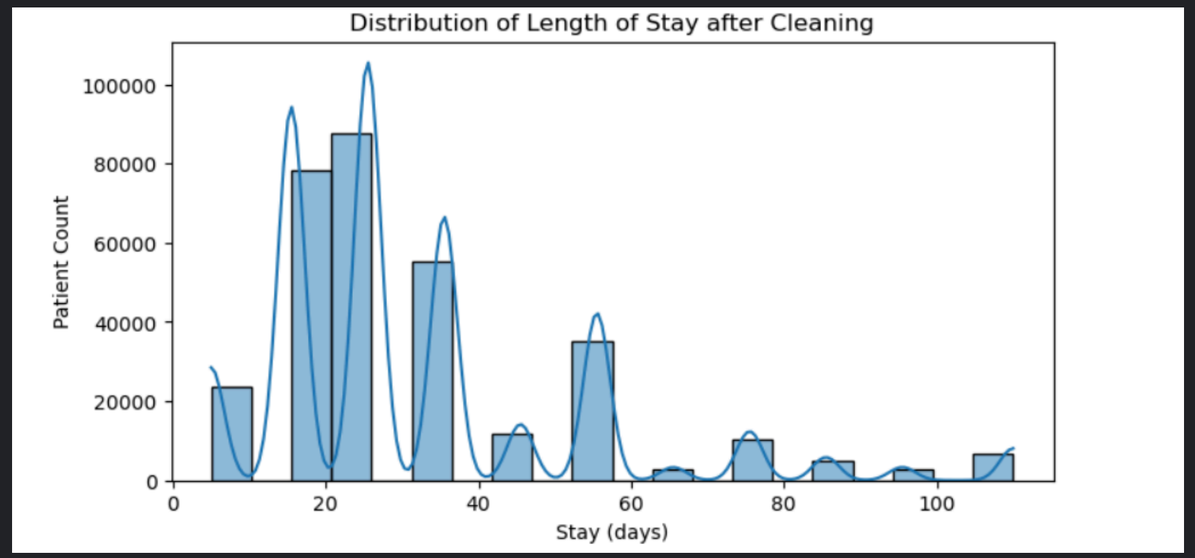
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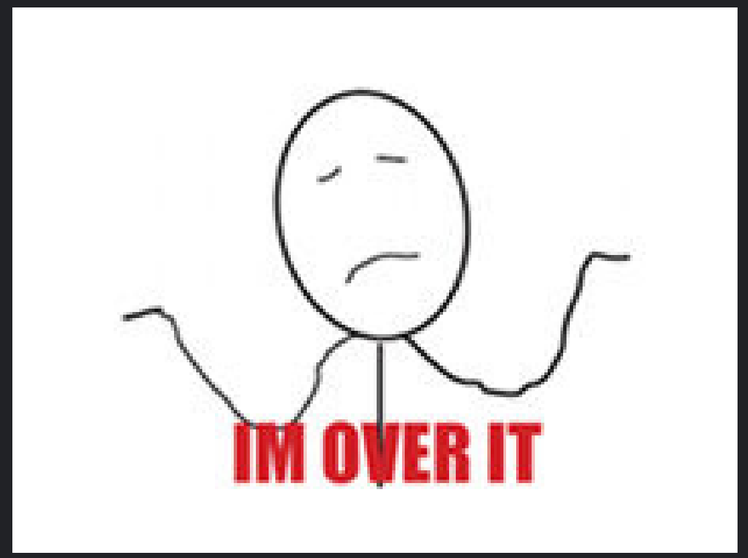
IM OVER IT

ANALYSIS MODEL

MODEL CREATION OF OUR RESEARCH QUESTION

→ Dependent Variable: Stay
→ Independent Variables: Visitors with Patient and Age

Visitors and Age values show mainly linear variance scales (instead of polynomial or logarithmic scales)
→ Linear Regression is the best suited model

COMPARING MSE RESULTS

Column	case_id	Hospital_code	Available Extra Rooms in Hospital	Visitors with Patient	Age	Visitors with Patient + Age
MSE	1.0095	1.0102	0.9986	0.7181	1.0033	0.7154

RESULTS

→ Visitors is the most prominent predictor in our data set

→ Including Age as an additional predictor improves the result (though just slightly)

STATSMODEL EVALUATION

RESULTS

MSE = 0.7154

R^2 = 0.290

p-values << 0.05

→ 'Visitors with Patient' and 'Age' are strong predictors for 'Stay'!

```python
target = "Stay"
predictors = ["Visitors with Patient", "Age"]
evalMethods = ["mean squared error", "StatsModel"]
X_train, X_test, y_train, y_test = prepareAnalysisData(df=scaledDataFrame, targetCol=target)
linearRegression(X_train=X_train, X_test=X_test, y_train=y_train, y_test=y_test, predictorsCol=predictors, evalMethods=evalMethods)
```

✓ 0.4s

```
-----------------------
Analysis:
Evaluation of ['Visitors with Patient', 'Age']according to mean squared error
Mean squared error:
0.7154730763920658
Evaluation of ['Visitors with Patient', 'Age']according to StatsModel
StatsModel Analysis:
                            OLS Regression Results
==============================================================================
Dep. Variable:                   Stay   R-squared:                       0.290
Model:                            OLS   Adj. R-squared:                  0.290
Method:                 Least Squares   F-statistic:                 5.212e+04
Date:                Wed, 23 Jul 2025   Prob (F-statistic):               0.00
Time:                        18:00:20   Log-Likelihood:            -3.1728e+05
No. Observations:              254660   AIC:                         6.346e+05
Df Residuals:                  254657   BIC:                         6.346e+05
Df Model:                           2
Covariance Type:            nonrobust
==========================================================================================
                            coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    -0.0015      0.002     -0.922      0.356      -0.005       0.002
Visitors with Patient     0.5284      0.002    317.843      0.000       0.525       0.532
Age                       0.0556      0.002     33.282      0.000       0.052       0.059
==============================================================================
Omnibus:                    41789.139   Durbin-Watson:                   1.997
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            84100.003
Skew:                           1.001   Prob(JB):                         0.00
Kurtosis:                       4.980   Cond. No.                         1.08
==============================================================================
```

Other Testing Methods Used : -

```
grouped = df.groupby('Department').mean()
print(grouped)
```

Aggreation and grouping

```
df = pd.DataFrame({'A': [1, 2, np.nan], 'B': [4, np.nan, 6]})
print("Original:\n", df)
df_filled = df.fillna(0)
print("After Fill:\n", df_filled)
```

Built-in Methods and Functions

CONCLUSION/SUMMARY

- We transformed a messy, real-world healthcare dataset into a clean, structured foundation.
- A majority of patients had short stays (under 30 days).
- Moderate ill patients and certain department (like gynecology) were associated with longer LOS.
- Key predictors such as Visitors with Patient and Age showed strong correlation with stay duration.
- Visualizations and statistical tests confirmed that our data-driven insights reflect real trends in patient stays.
- Our insights can help hospitals optimize beds, staffing, and discharge planning.

 # Well-structured data leads to smarter decisions in healthcare.

THANK YOU FOR LISTENING

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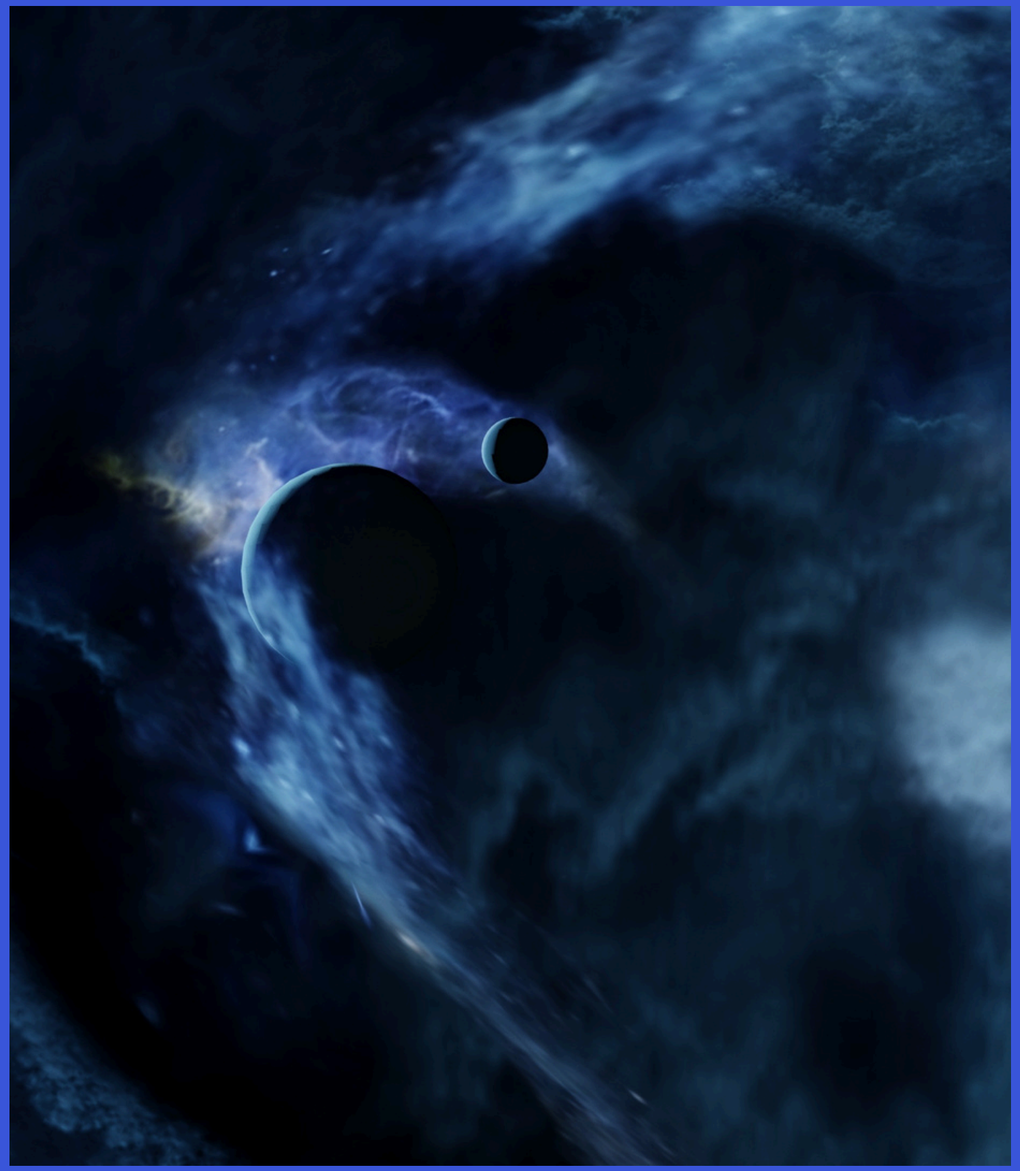
Maximilian von Quast
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Raghvendra Singh